L17 Cauchy-Euler Equation and Reduction of Order

Section 4.7

We have seen that a nonhomogeneous constant-coefficient differential equation

$$ay'' + by' + cy = f(t),$$

constants

has solutions valid over all intervals where $f(t)$ is continuous (ensuring that the integrals containing $f(t)$ exist and are differentiable). But what meaning does the differential equation have at a point where $f(t)$ is undefined, or discontinuous?

When we have equations with variable coefficients of the form

$$a_2(t)y'' + a_1(t)y' + a_0(t)y = f(t) \ ,$$

the most we can expect is that there are solutions that are valid over intervals where all four functions, $a_2(t)$, $a_1(t)$, $a_1(t)$, and $f(t)$, are continuous. The coefficient function $a_2(t)$ must be nonzero over the interval.

$a_2 \neq 0$

standard form: $y'' + \underbrace{\frac{a_1}{a_2}}_{p} y' + \underbrace{\frac{a_0}{a_2}}_{q} y = \underbrace{\frac{f(t)}{a_2}}_{g}$

Existence and Uniqueness of Solutions Theorem.

Suppose $p(t), q(t)$, and $g(t)$ are continuous on an interval (a,b) that contains the point t_0. Then, for any choice of the initial values Y_0 and Y_1, there exists a unique solution $y(t)$ on the same interval (a,b) to the initial value problem

$$\text{IVP} \begin{cases} y''(t) + p(t)y'(t) + q(t)y(t) = g(t); \\ y(t_0) = Y_0,\ y'(t_0) = Y_1\ . \end{cases}$$

ex. Determine the largest interval for which the theorem of existence and uniqueness ensures the existence and uniqueness of a solution to the initial value problem.

at $t_0 = 1$

$$\underbrace{\overset{a_2}{(t-3)}y'' + \overset{a_1=1}{y'} + \overset{a_0}{\sqrt{t}}y = \overset{f}{\ln t},}\quad y(1) = 3,\ y'(1) = -5$$

standard form: $y'' + \frac{1}{t-3}y' + \frac{\sqrt{t}}{t-3}y = \frac{\ln(t)}{t-3}$

$p(t) = \frac{1}{t-3} \to$ cont: $(-\infty, 3) \cup (3, \infty)$

$q(t) = \frac{\sqrt{t}}{t-3} \to$ cont: $[0,3) \cup (3, \infty)$

$g(t) = \frac{\ln(t)}{t-3} \to$ cont: $(0,3) \cup (3, \infty)$

all p, q, g cont on $(0,3) \cup (3, \infty)$

$\uparrow$ 1

guaranteed a unique solution on $(0,3)$ for $t_0 = 1$.

Cauchy-Euler Equations

In this lecture, we consider the linear second-order equation with variable coefficients of the form

$$a(t)y'' + b(t)y' + c(t)y = f(t)$$

Def. A linear equation of the form

powers align with order of derivative.

$$a_n t^n \frac{d^n y}{dt^n} + a_{n-1} t^{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 t \frac{dy}{dt} + a_0 y = f(t),$$

where $a_n, a_{n-1}, \cdots, a_0$ are constants, is known as a **Cauchy-Euler equation**. Its characteristic is that the degree, k, of the coefficients t^k matches the order k of the differentiation $d^k y/dt^k$.

or equidimensional.

ex. Determine if the equation is Cauchy-Euler equation.

yes!

1. $3t^2y'' + 11ty' - 3y = \sin t$ Cauchy-Euler.

2. $2y'' - 3ty' + 11y = 3t - 1$ not Cauchy-Euler

   not t^2

3. $3ty'' + 11y' = t^4$

   $\Rightarrow 3t^2y'' + 11ty' = t^5$ Yes, Cauchy-Euler.

$$\overset{y}{t^n} \xrightarrow{d} \overset{y'}{n t^{n-1}}$$

$$t \cdot n t^{n-1} = n t^n$$

First, consider a homogeneous Cauchy-Euler equation:

$$at^2y'' + bty' + cy = 0 \quad (t > 0) \qquad (*)$$

$$\Rightarrow y'' + \frac{b}{at} + \frac{c}{at^2} = 0 \quad t \neq 0$$

cont $(-\infty, 0) \cup (0, \infty)$

Try solutions of the form $y = t^r$

(since each term will have the form $\underline{(\text{constant}) \times t^r}$:)

Then $\quad ty' = t \cdot (r t^{r-1}) = r t^r$

$$t^2y'' = t^2 (r(r-1) t^{r-2}) = r(r-1) t^r$$

Solve for r by plugging into the homogeneous equation (*):

$$\Longrightarrow \quad a(\overset{t^2y''}{r(r-1)t^r}) + b\overset{ty'}{r t^r} + c\overset{y}{t^r} \underset{\text{want}}{=} 0$$

$$\Rightarrow \quad t^r[a(r)(r-1) + br + c] = 0$$

$t \neq 0$ $\quad \Rightarrow ar^2 - ar + br + c = 0$

which gives $\quad ar^2 + (b-a)r + c = 0,$

the **characteristic equation** for the Cauchy-Euler equation. Thus, $y = t^r$ is a solution of the DE whenever r is a solution of the characteristic equation.

There are 3 different cases to be considered, depending on whether the roots r are real & distinct, real & equal, or complex.

Method of finding two independent solution to the homogeneous Cauchy-Euler equation:

Given a 2nd-order homogeneous Cauchy-Euler equation

$$at^2y'' + bty' + cy = 0,$$

the associated auxiliary equation is

$$\underline{ar^2 + (b-a)r + c = 0.} \rightarrow r = \frac{-(b-a) \pm \sqrt{(b-a)^2 - 4ac}}{2a}$$

<u>Case 1:</u> two distinct real roots $(r_1 \neq r_2)$ $\quad (b-a)^2 - 4ac > 0$

The general solution is $y_h(t) = c_1 t^{r_1} + c_2 t^{r_2}$. $\rightarrow W = \begin{vmatrix} t^{r_1} & t^{r_2} \\ r_1 t^{r_1-1} & r_2 t^{r_2-1} \end{vmatrix}$

$= r_2 t^{r_1+r_2-1} - r_1 t^{r_2+r_1-1}$

$= \underset{\neq 0}{(r_2 - r_1)} \underset{t \neq 0}{t^{r_2+r_1-1}}$

<u>Case 2:</u> one repeated real root (r)

The general solution is $y_h(t) = c_1 t^r + c_2 t^r \underline{\ln t}$.

(this result arises from <u>*Reduction of Order*</u>)

<u>Case 3:</u> two conjugate complex roots

$$(r = \alpha \pm i\beta,\ \beta \neq 0)$$

When $r = \alpha + i\beta$, we have $t^{\alpha+i\beta}$.

$t^{\alpha+i\beta} = \left(e^{\ln(t)}\right)^{\alpha+i\beta}$

$= e^{(\alpha+i\beta)\ln(t)}$

Use the fact that $t = e^{\ln t}$ along with Euler formula, $t^{\alpha+i\beta} = t^{\alpha} \cdot t^{i\beta}$

$$= t^{\alpha} \cdot e^{(\ln t)^{i\beta}},$$

$$= t^{\alpha} \cdot e^{i[\beta(\ln t)]},$$

$$= t^{\alpha}[\cos(\beta \ln t) + i \sin(\beta \ln t)]$$

Simplifying the same way as we did in section 4.3, we get 2 linearly independent REAL solutions:

$$y_1 = t^\alpha(\cos(\beta \ln t)), \qquad y_2 = t^\alpha(\sin(\beta \ln t)).$$

The general solution is

$$y_h(t) = c_1 t^\alpha \cos(\beta \ln t) + c_2 t^\alpha \sin(\beta \ln t).$$

<u>ex.</u> Find two linearly independent solutions to the equation:

$ar^2 + (b-a)r + c = 0$

1. $t^2y'' + 3ty' - 3y = 0$

$a=1,\ b=3,\ c=-3 \Rightarrow$ char eqn: $1 \cdot r^2 + (3-1)r - 3 = 0$

$\Rightarrow r^2 + 2r - 3 = 0 \Rightarrow (r+3)(r-1) = 0$

$\Rightarrow r = 1, -3$

$y_h = c_1 t^1 + c_2 t^{-3}$

2. $t^2y'' + 3ty' + y = 0$

$a=1,\ b=3,\ c=1 \Rightarrow$ char eqn: $r^2 + 2r + 1 = 0$

$(r+1)^2 = 0$

$r = -1, -1$

$y_h = c_1 t^{-1} + c_2 t^{-1} \ln(t)$

3. $3t^2y'' + ty' + 2y = 0$

$a=3,\ b=1,\ c=2 \Rightarrow$ char eqn: $3r^2 + (-2)r + 2 = 0$

$\Rightarrow 3r^2 - 2r + 2 = 0$

$r = \dfrac{2 \pm \sqrt{4 - 4(3)(2)}}{6}$

$= \dfrac{2 \pm \sqrt{-20}}{6} = \underbrace{\tfrac{1}{3}}_{\alpha} \pm \underbrace{\tfrac{\sqrt{5}}{3}}_{\beta} i$

$$\boxed{y_h = c_1 t^{1/3} \cos\left(\tfrac{\sqrt{5}}{3} \ln(t)\right) + c_2 t^{1/3} \sin\left(\tfrac{\sqrt{5}}{3} \ln(t)\right)}$$

For the non-homogeneous equation

$$at^2y''(t) + bty'(t) + cy(t) = f(t),$$

a general solution is given by

$$y_g(t) = y_p(t) + y_h(t)$$

where y_p is a particular solution to the non-homogeneous equation and y_h is a general solution to the corresponding homogeneous equation.

Solution method finding a general solution to the non-homogeneous Cauchy-Euler equation:

• Since the DE is non-homo, we first solve for associated homogeneous equations using its characteristic equation

$$ar^2 + (b-a)r + c = 0$$

to find 2 linearly independent solutions y_1, y_2.
• To find a particular solution, we use **variation of parameters**. We compute W, W_1, W_2 as defined before to solve for v_1', v_2'.

$ar^2 + (b-a)r + c = 0$

ex. Find a general solution.

$$t^2y'' - 3ty' + 3y = 2t^4e^t \quad (t > 0)$$

$a = 1$, $b = -3$, $c = 3 \Rightarrow$ char eqn: $r^2 + (-3-1)r + 3 = 0$

$r^2 - 4r + 3 = 0$

$(r-3)(r-1) = 0$

$r = 1, 3$

$$y_h = c_1 \overset{y_1}{t} + c_2 \overset{y_2}{t^3}$$

$$y_1 = t,\ y_2 = t^3, \quad \frac{f}{a} = \frac{2t^4e^t}{t^2} = 2t^2e^t$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t & t^3 \\ 1 & 3t^2 \end{vmatrix} = 3t^3 - t^3 = 2t^3$$

IBP

$$\begin{array}{ccc} t^2 & & e^t \\ 2t & - & e^t \\ 2 & + & e^t \\ 0 & & e^t \end{array}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f/a & y_2' \end{vmatrix} = \begin{vmatrix} 0 & t^3 \\ 2t^2e^t & 3t^2 \end{vmatrix} = 0 - 2t^5e^t$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f/a \end{vmatrix} = \begin{vmatrix} t & 0 \\ 1 & 2t^2e^t \end{vmatrix} = 2t^3e^t - 0$$

$$v_1 = \int \frac{W_1}{W}\,dt = \int \frac{-2t^5e^t}{2t^3}\,dt = \int -t^2e^t\,dt = -(t^2e^t - 2te^t + 2e^t)$$

$$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{2t^3e^t}{2t^3}\,dt = \int e^t\,dt = e^t$$

$$y_g = y_h + y_p = (c_1t + c_2t^3) + \left[(\cancel{-t^2e^t} + 2te^t - 2e^t)t + \cancel{e^tt^3}\right]$$

ex. Find a general solution given two linearly independent homogeneous solutions to the corresponding homogeneous equation.

$$ty'' + (1-2t)y' + (t-1)y = te^t \quad t > 0$$

a $\qquad$ f

$$y_1(t) = e^t,\ y_2(t) = e^t \ln t$$

use variation of parameters: $\frac{f}{a} = \frac{te^t}{t} = e^t$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^t & e^t\ln(t) \\ e^t & e^t\ln(t) + \frac{e^t}{t} \end{vmatrix} = \left(e^{2t}\ln(t) + \frac{1}{t}e^{2t}\right) - e^{2t}\ln(t)$$

$$= \frac{1}{t}e^{2t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f/a & y_2' \end{vmatrix} = \begin{vmatrix} 0 & e^t\ln(t) \\ e^t & e^t\ln(t) + \frac{e^t}{t} \end{vmatrix} = 0 - e^{2t}\ln(t)$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f/a \end{vmatrix} = \begin{vmatrix} e^t & 0 \\ e^t & e^t \end{vmatrix} = e^{2t} - 0$$

IBP

$\ln(t) \quad -t$

$\frac{1}{t} \rightarrow -\frac{1}{2}t^2$

$$v_1 = \int \frac{W_1}{W}\,dt = \int \frac{-e^{2t}\ln(t)}{\frac{1}{t}e^{2t}}\,dt = \int -t\ln(t)\,dt$$

$$= -\frac{1}{2}t^2\ln(t) + \int \frac{1}{2}t\,dt$$

$$= -\frac{1}{2}t^2\ln(t) + \frac{1}{4}t^2$$

$$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{e^{2t}}{\frac{1}{t}e^{2t}}\,dt = \int t\,dt$$

$$= \frac{1}{2}t^2$$

So $y_g = y_h + y_p = \left(c_1 e^t + c_2 e^t\ln(t)\right) + \left(\overset{v_1}{\cancel{\left(-\frac{1}{2}t^2\ln(t)\right)}} + \underbrace{\frac{1}{4}t^2}_{v_2}\right)\overset{y_1}{e^t}$

$$+ \underbrace{\cancel{\frac{1}{2}t^2 e^t\ln(t)}}_{v_2\ y_2}\Big)$$

Theorem. (Reduction of Order)

Let $y_1(t)$ be a solution (not identically zero) to equation

(*) $y'' + p(t)y' + q(t)y = 0$ in an interval I. Then

$$y_2(t) = y_1(t) \int \frac{e^{-\int p(t)dt}}{y_1^2(t)} dt$$

is a second linearly independent solution.

as y_1 solves (*)

$\underline{y_1'' + p y_1' + q y_1 = 0}$

Proof. Set $y_2 = v y_1 \Rightarrow y_2' = v' y_1 + v y_1'$

$y_2'' = v'' y_1 + 2v' y_1' + v y_1''$

$$y_2'' + p y_2' + q y_2 = (v'' y_1 + 2v' y_1' + v y_1'') + p(v' y_1 + v y_1') + q(v y_1)$$

$$= v'' y_1 + 2v' y_1' + p v' y_1 + \underbrace{[v(y_1'' + p y_1' + q y_1)]}_{=0} \underset{\text{want}}{=} 0$$

$\Rightarrow y_1 v'' + (2y_1' + p y_1) v' = 0$ set $w = v'$

$\Rightarrow y_1 w' + (2y_1' + p y_1) w = 0 \Rightarrow w' + \left(\frac{2y_1' + p y_1}{y_1}\right) w = 0$

$\Rightarrow \mu = e^{\int \frac{2y_1' + p y_1}{y_1} dt} = e^{\int 2\frac{y_1'}{y_1} + p\, dt}$

$u = y_1$

$du = y_1' dt$

$= e^{\int 2 \cdot \frac{1}{u} dt + \int p dt} = e^{2\ln|u|} e^{\int p dt} = y_1^2 e^{\int p dt}$

$w = \frac{1}{\mu}[\int \mu \cdot 0 + C] = \frac{C}{\mu} = C \cdot y_1^{-2} e^{-\int p dt}$

$$v' = c \cdot \frac{e^{-\int p\,dt}}{y_1^2} \Rightarrow v = \int c\,\frac{e^{-\int p\,dt}}{y_1^2}\,dt$$

So $$y_2 = v y_1 = y_1 \cdot \int \cancel{c}\,\frac{e^{-\int p\,dt}}{y_1^2}\,dt \quad \square$$

$r^2 - 4r + 4 = 0 \quad r = 2, 2$

$(r-2)^2 = 0 \quad e^{2t}, te^{2t}$

ex. Given that $y_1 = e^{2t}$ is a solution to $y'' - 4y' + 4y = 0$, find a second linearly independent solution.

$p(t) = -4$

?

reduction of order: $y_2 = y_1 \int \frac{e^{-\int p\,dt}}{y_1^2}\,dt$

$$\Rightarrow y_2 = e^{2t} \int \frac{e^{-\int -4\,dt}}{(e^{2t})^2}\,dt = e^{2t} \int \frac{e^{\int 4\,dt}}{e^{4t}}\,dt$$

$$= e^{2t} \int \frac{e^{4t}}{e^{4t}}\,dt = e^{2t} \int dt = te^{2t} \checkmark$$

ex. Find a second linearly independent solution. $t^2y'' - ty' + y = 0$

$$y'' - \frac{1}{t}y' + \frac{1}{t^2}y = 0,\ t > 0,\ y_1(t) = t$$

C-E

$p(t)$

reduction of order : $y_2 = y_1 \int \frac{e^{-\int p(t)dt}}{y_1^2}\, dt$

$$\Rightarrow y_2 = t\int \frac{e^{-\int -\frac{1}{t}dt}}{t^2}\, dt = t\int \frac{e^{\int \frac{1}{t}dt}}{t^2}\, dt = t\int \frac{e^{\ln(t)}}{t^2}\, dt$$

$$= t\int \frac{t}{t^2}\, dt = t\int \frac{1}{t}\, dt = t\ln(t)$$

$t^2y'' - ty' + y = 0$

$a=1,\ b=-1,\ c=1,\ ar^2 + (b-a)r + c = 0$

$r^2 + (-1-1)r + 1 = 0 \Rightarrow r^2 - 2r + 1 = 0$

$(r-1)^2 = 0$

$r = 1, 1$

ex. Find a second linearly independent solution.

$$ty'' + (1-2t)y' + (t-1)y = 0,\ t > 0,\ y_1(t) = e^t$$

$$\Rightarrow y'' + \underbrace{\frac{1-2t}{t}}_{p(t)} y' + \frac{t-1}{t} y = 0$$

So $$y_2 = y_1 \int \frac{e^{-\int p(t)dt}}{y_1^2}\, dt = e^t \int \frac{e^{-\int \frac{1-2t}{t} dt}}{(e^t)^2}\, dt$$

$$\Rightarrow y_2 = e^t \int \frac{e^{\int -\frac{1}{t} + 2\, dt}}{e^{2t}}\, dt = e^t \int \frac{e^{-\ln(t)} \cancel{e^{2t}}}{\cancel{e^{2t}}}\, dt$$

$$= e^t \int t^{-1} dt = e^t \ln(t)$$

Extra: Abel's Formula

$$W[y_1,y_2](t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'$$

$$\boxed{\left(\frac{y_2}{y_1}\right)' = \frac{y_1y_2' - y_2y_1'}{(y_1)^2}}$$

Suppose y_1, y_2 solve $y'' + py' + qy = 0$ $\quad y'' + py' = -qy$

$$W' = (y_1y_2' - y_2y_1')' = \cancel{y_1'y_2'} + y_1y_2'' - (\cancel{y_2'y_1'} + y_2y_1'')$$
$$= y_1y_2'' - y_2y_1''$$

Look at $W' + pW = (y_1y_2'' - y_2y_1'') + p(y_1y_2' - y_2y_1')$

$$= y_1(\underbrace{y_2'' + py_2'}) - y_2(\underbrace{y_1'' + py_1'})$$

$$= y_1(-qy_2) - y_2(-qy_1) = 0$$

So W satisfies $W' + pW = 0$ solve

$$\mu = e^{\int p(t)dt}$$

$$W = \frac{1}{\mu}\left[\int \mu \cdot 0\, dt + C\right] = C \cdot \frac{1}{\mu} = C \cdot e^{-\int p(t)dt}$$

written with an interval (a,b) choose, t_0, t.

Abel's Formula

$$W = Ce^{-\int_{t_0}^{t} p(v)dv} \quad \text{for any } t_0, t \text{ in } (a,b)$$

$W \neq 0 \Rightarrow C \neq 0$ for any t_0 in (a,b)

$$W(t_0) = y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)$$

<u>Reduction of Order</u>: If y_1, y_2 satisfy $y'' + py' + qy = 0$

then $$\left(\frac{y_2}{y_1}\right)' = \frac{y_1y_2' - y_2y_1'}{y_1^2} = \frac{W[y_1,y_2]}{y_1^2} = \frac{Ce^{-\int p(t)dt}}{y_1^2}$$

Abel's Formula

$$\Rightarrow \frac{y_2}{\cancel{y_1}} = y_1\int\left(\frac{y_2}{y_1}\right)'dt = y_1\int\frac{Ce^{-\int p(t)dt}}{y_1^2}dt$$

<u>Construct Some Equations</u>

Given y_1, y_2 LI then we can construct

$$\begin{vmatrix} y_1 & y_2 & y \\ y_1' & y_2' & y' \\ y_1'' & y_2'' & y'' \end{vmatrix} = \begin{vmatrix} y_1' & y_2' \\ y_1'' & y_2'' \end{vmatrix} y - \begin{vmatrix} y_1 & y_2 \\ y_1'' & y_2'' \end{vmatrix} y' + \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} y'' \underset{\text{want}}{=} 0$$

$$\Rightarrow (y_1'y_2'' - y_1''y_2')y - (y_1y_2'' - y_1''y_2)y' + (y_1y_2' - y_1'y_2)y'' = 0$$

important W

$$\Rightarrow \underbrace{(y_1y_2' - y_1'y_2)}_{\neq 0}y'' + (y_1''y_2 - y_1y_2'')y' + (y_1'y_2'' - y_1''y_2')y = 0$$

y_1, y_2 both satisfy this equation!

Ex: $y_1 = t$, $y_2 = \tan(t)$ $\quad y_1' = 1 \quad y_2' = \sec^2(t)$

$y_1'' = 0 \quad y_2'' = 2\sec^2(t)\tan(t)$

$$(t\sec^2(t) - \sec^2(t))y'' + (0 - 2t\sec^2(t)\tan(t))y' + (2\sec^2(t)\tan(t) - 0)y = 0$$

$$\Rightarrow \sec^2(t)(t-1)y'' - 2t\sec^2(t)\tan(t)y' + 2\sec^2\tan(t)y = 0$$

$$\Rightarrow (t-1)y'' - 2t\tan(t)y' + 2\tan(t)y = 0.$$